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Original Research Paper

Road pavement condition diagnostics using smartphone-based data crowdsourcing in smart cities



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HIGHLIGHTS

- Using smartphone-based data crowd sourcing for road pavement condition assessment.
- Identifying road pavement condition based on analysis of linear accelerations.
- Studying the intervention-free measurements by transport system users.
- Using binary classifiers to assess the efficiency of road infrastructure diagnostics.

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ABSTRACT

The purpose of the paper is to analyse the effectiveness of a solution known as road condition tool (RCT) based on data crowdsourcing from smartphones users in the transport system. The tool developed by the author of the paper, enabling identification and assessment of road pavement defects by analysing the dynamics of vehicle motion in the road network. Transport system users equipped with a smartphone with the RCT mobile application on board record data of linear accelerations, speed, and vehicle location, and then, without any intervention, send them to the RCT server database in an aggregated form. The aggregated data are processed in the combined time and location criterion, and the road pavement condition assessment index is estimated for fixed 10 m long measuring sections. The measuring sections correspond to the sections of roads defined in the pavement management systems (PMS) used by municipal road infrastructure administration bodies. Both the research in question and the results obtained by the method proposed for purposes of the road pavement condition assessment were compared with a set of reference data of the road infrastructure administration body which conducted surveys using highly specialised measuring equipment. The results of this comparison, performed using binary classifiers, confirm the potential RCT solution proposed by the author. This solution makes it possible to global monitor the road infrastructure condition on a continuous basis via numerous users of the transport system, which guarantees that such an assessment is kept up to date.

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1. Introduction

A smart city is a kind of space where information and communication technologies (ICT) are used to improve acquisition, distribution and utilisation of global data in people's environment, and to enhance the interaction between urban infrastructure subsystems as well as their functionality. Implementation of technologically advanced solutions, such as sensors, mobile solutions or embedded systems, makes it possible to create a digital dimension of such space (Amaxilatis et al., 2018). The notion of a smart city also stands for investing in human capital in order to actively foster and mould sustainable development of economic growth, transport and adequate management of natural resources with the support of the society. Creating and developing the idea of space as a smart city by local authorities is assumed to consequently bring benefits to city inhabitants, entities operating in the public sector, the industry as well as service providers (Baltac, 2019).

What matters particularly at the stage of the smart city space building is organising the resources at hand, updating the available data, and acquiring new data using state-of-the-art technologies, the latter to be used and supervised by local authorities, but also rendered to the inhabitants and businesses striving to fulfil their goals in the urban space (Staniek, 2018b). By definition, the smart city concept also encompasses the transport system as well as the pursuit of its efficient use by ensuring access, performance, safety and travelling comfort to its users.

The transport system, when perceived by the travelling population as a source of nuisance preventing convenient mobility due to its insufficiently developed infrastructure and advancing degradation, may also be a critical component of the smart city idea implementation. Poor technical condition of pavement as well as structural defects of roads reduce traffic safety and comfort for traffic participants, but they also exert a negative impact on the operation of automotive vehicles. By continuously monitoring the condition of road pavement, one can establish how long the given road infrastructure element can remain in service without any symptoms of defect, and consequently prevent further degradation should specific symptoms be discovered. A defect description thus obtained determines whether or not specific ad-hoc or major repairs are to be performed, and if so, to what extent. Using such technologies as microelectromechanical system (MEMS), global positioning system (GPS) and global system for mobile (GSM) with mobile devices, as well as mapping measurement results in the geographic information system (GIS) can provide considerable support for the pavement management system, also making it possible to create the digital space dimension which proves to be crucial for the smart city idea.

2. Related works

Visual inspection is the most popular road pavement condition monitoring method, and it is typically performed by qualified and seasoned inspectors who identify and survey

road pavement defects. The inspections are carried out directly on roads, often under conditions of high traffic intensity and high vehicle running speed. Despite being popular, this method has many drawbacks (Mustaffara et al., 2008). Some alternatives to this solution include vision methods used to detect road defects as well as methods of analysis of linear accelerations for vehicle shocks which characterise road pavement irregularities.

The author's literature review pertaining to the problems of road pavement condition assessment encompass a group of solutions based on the analysis of linear accelerations of vehicles moving in the road network. The use of mobile devices, namely smartphones, as an element of the internet of things vision for purposes of the road infrastructure condition monitoring has been described in general terms (Ghose et al., 2012). Du et al. (2016) proposed a more detailed approach to this problem addressing an integrated measuring system comprising a data collector, an on-board car terminal, and a processing system for the data previously acquired. Field tests were conducted in selected sections of trunk roads in Huzhou, and the results thus obtained were compared with the real-life RQI, verifying the accuracy, performance and stability of the measuring tool designed under the research project. The maximum relative error of the solution proposed was below 10%.

Under the research referred to in Singh et al. (2017), the smart-patrolling system was proposed to be used in the roads of the city of Chandigarh to assess the condition of road pavement. This solution is based on a combination of measurement data from multiple streams, namely smartphones using crowdsourcing, followed by an analysis of the data obtained by dynamic time warping (DTW). Compared to other similar methods, the one proposed manages well under variable traffic conditions and at different speeds. The advantage of the DTW solution is the simplicity of its implementation as well as the speed of calculation compared to support vector machine (SVM), hidden Markov model (HMM), and artificial neural network (ANN). The average detection rate of the proposed solution is 88.66%. Lima et al. (2017) comprise a mobile application running on the Android operating system, known as road-scan, used for real-time recording of measurement data, and for the crowdsourcing service enabling the road pavement condition to be assessed. The four assessment classes defined in this case are conducive to accuracy of the road infrastructure condition description. The research implies that it is possible to identify the condition by means of accelerometers, and the use of the data provided by many smartphone users helps in reducing the error rate of the road pavement assessment. An analogical solution based on the use of the Asfalt mobile application and the crowdsourcing service to aggregate data and monitor the condition of asphalt pavement has been discussed (Souza et al., 2018). On account of the limited precision of a smartphone accelerometer compared to professional road profiling equipment, the authors used an approach based on machine learning and signal processing techniques. Moreover, they proposed a dynamic model capable of assessing the road pavement condition in a four-level scale (good, average, fair, and poor), as well as pavement defects. Asfalt is

characterised by an accuracy of 90% under various operating conditions, however, measurements performed using the Asfalt system require a sampling frequency of at least 100 Hz. The problem of applying the machine learning algorithm to forecast the road infrastructure condition was analysed by [Allouch et al. \(2017\)](#), based on the use of an accelerometer and a gyroscope for data acquisition as well as a GPS module for route plotting on a map. The authors analysed three classification algorithms: the C4.5 decision tree, SVM, and naive Bayes. The research revealed superiority of C4.5 in terms of detection accuracy (98.6%), taking two measurement sensors into account, i.e., an accelerometer and a gyroscope installed in smartphones. Moreover, the advantage of this solution, i.e., the road-sense application, is providing the users with knowledge on the condition of transport routes. [Xue et al. \(2017\)](#) address a study of the problem of perception of a pothole profile in a metropolitan area. What was developed under the study in question is the P3 application which uses smart learning algorithms to identify different categories of vehicles in the process of measurement and estimation of the road profile including damage detection. The P3 application ensures high efficiency of road description provided that the vehicle speed during measurements is low and the sampling frequency high. The field tests confirmed that the P3 application can achieve a low level of error in defect depth estimation, i.e., 13%, as well as the defect length estimation error of 16%.

A fuzzy inference system used for detecting uneven road pavements was proposed by [Aljaafreh et al. \(2017\)](#). This solution is based on the principle of variance of vertical acceleration and vehicle speed, as recorded with a smartphone. The related tests were performed at different speeds, and the results obtained show that the proposed method is promising considering detection of bumps. The proposed fuzzy inference system combines the effect of vibrations and vehicle speed to effectively eliminate the signal noise. A disadvantage of this solution is that it reduces the system's efficiency for high vehicle speeds and entails the need to fine tune the fuzzy logic parameters, especially in high speed ranges. A similar solution used for fuzzy logic-based classification of road pavement irregularities was presented by [Kumar et al. \(2017\)](#), and the related tests of its operating efficiency were conducted in the road network of the northern regions of India.

[Agostinacchio et al. \(2014\)](#) address the problem of a dynamic effect of the loads transferred by vehicles on the road pavement in specific technical condition. In accordance with the applicable standards (ISO 8608), the authors determined the road roughness profiles, while using the MATLAB application, they simulated a quarter-car model against pre-set speed intervals. Their studies have shown that vehicle vibrations essentially depend on the road's longitudinal evenness, and only to a small degree on the changes to the vehicle running speed. The on-board car terminal has been designed in such a manner as to enable data recording and transfer following integration of linear acceleration related data, GPS location data, and information on the driving style.

With a large-scale assessment of the road network condition in mind, aiming at high accuracy and precision of the

measuring system, the solution proposed by [Zhao et al. \(2016\)](#) is the dynamic response intelligent monitoring system (iDRIMS) based on using a typical smartphone running on iOS. The iDRIMS add-on described demonstrated in [Zhao and Nagayama \(2017\)](#) makes it possible to assess the road condition from the international roughness index (IRI) perspective on the basis of the linear accelerations, angular velocity, and GPS location data previously recorded, with high-precision sampling time. The results of both the numerical simulation and the experiment imply that the method proposed enables accurate determination of the given road's IRI with reference to vehicles of different categories, any chosen arrangement of sensors on board the vehicle, as well as different running speeds when measuring.

What was proposed in [Li and Goldberg \(2018\)](#) is a low-budget system for the road pavement condition assessment based on built-in sensors of popular smartphones. Its operational analysis was verified with reference to two descriptive indicators: IRI-proxy, and the number of recorded transient events. The solution was also verified using test data received from five system users, recorded in routes of circle around 50 km travelled around College Station, Texas. The relevant field tests showed that built-in sensors of commonly available smartphones combined with the proposed measurement method are effective in road quality assessment and detection of transient events.

[Yi et al. \(2015\)](#) proposed another system intended for road pavement monitoring, known as smartphone-probe-car. It allows for detection and indexing of road defects, and on top of that, it is also an adaptive system enabling selection of the best parameters for an algorithm which adjusts itself to the variable speed of the test vehicle. This means 44 similar values of road defect assessment indicators at different vehicle speeds. Moreover, in order to demonstrate the benefits of improving detection accuracy at the system level, data clustering based on a pre-defined measurement step was used.

[Harikrishnan and Gopi \(2017\)](#) provide a proposal of a method for road pavement monitoring, pothole and bump detection, as well as prediction of their consequences by analysing signals of vertical vibration of vehicles in traffic in the road network. The system thus developed uses a smartphone accelerometer to record vehicle vibrations in the vertical axis. The defect identification algorithm is based on the Gaussian model, while x-z ratio filtering is used for defect classification as either a pothole or a bump. Additionally, the paper proposes an algorithm for estimating the consequences of road defects per a vehicle with reference to the relationship between vertical acceleration and relative vertical displacement of the vehicle, provided that the running speed of the test vehicle ranges between 15 and 20 km/h.

A comparison of the effectiveness of the Nottingham pavement infrastructure defect detection algorithm (NPIDDA) defect identification system using a smartphone accelerometer with a method of visual assessment of road infrastructure condition was provided by [Byrne et al. \(2013\)](#). The related damage detection algorithm is based on multiple measurements conducted at different speeds and with the sampling frequency of 10 Hz. The field test results match the

description of the road infrastructure, while a network application enables continuous mapping of the network condition in real time. The application's key components are the data filtering algorithm based on identification of sudden events, such as emergency braking or vehicle tilting at sharp bends, and cut-off thresholds determined by the vehicle traffic dynamics resulting from the driver's driving style, the vehicle category, the suspension system type and condition, as well as the road pavement type.

Seraj et al. (2016) provide a discussion on application of stationary wavelet analysis and methods of elimination of changes in vehicle motion characteristics using the envelope technique for the sake of assessment of road irregularities. In order to evaluate the proposed solution, visual marking techniques were used, demonstrating a constant accuracy of 90% in detecting serious road pavement defects regardless of the vehicle type and location on the road.

Zang et al. (2018), on the other hand, focus on the assessment of irregularities of bicycle and pedestrian road pavements. The method in question is based on estimation of the IRI coefficient with reference to data retrieved from the GPS and MEME modules of a smartphone mounted on a bicycle. The method verification provided in the paper implies that the measurement results are heavily and positively correlated with the values measured using professional instruments. Additionally, the method proposed in the article makes it possible to identify potholes and bumps, and pinpoint their locations when the relevant measurements are performed by means of a bicycle. It should be noted that such an approach is particularly useful in the case of roads which are inaccessible to advanced measuring instruments, and these include footpaths and bike paths. Using the crowdsourcing capabilities, irregularities of pavement on footpaths and bike paths can be widely identified and presented to the users of two-wheelers.

Douangphachanh et al. (2013) comprised investigations pertaining to the characteristics and relationships between the linear acceleration data recorded by smartphones and the status of road pavement irregularities. The data collected in the measurements were divided into 100 m sections, and analysed in the frequency domain in order to calculate the values which characterise the accelerations thus obtained. The related analysis implied a linear relationship between the recorded linear accelerations and the road pavement condition. However, the significance of the relationship depends on the speed during measurements, which should be lower than 60 km/h. Moreover, it was proved that the vehicle type and the device type, i.e., a smartphone, affect the relationship between the recorded linear accelerations of the vehicle and the road pavement irregularity assessment index.

Results of another study (Douangphachanh and Oneyama, 2014) confirm that the road irregularity condition is correlated with the linear function of acceleration values and the mean vehicle speed. What has also been evidenced in this paper is that a linear acceleration signal corresponding to road pavement irregularity is the best measure to characterise the road pavement condition within the data reading and logging frequency range of 40–50 Hz.

The possibility of testing numerous solutions based on the function of an accelerometer built into a smartphone was presented by Carlos et al. (2018). The authors developed a platform for testing different algorithms of road pavement condition assessment, as described in the literature of the subject. Using the platform, they created a set of data concerning 30 roads with defects such as potholes and bumps, which they used to evaluate the heuristics also described in the literature. Considering the reference methodologies, the best results were obtained in the study of the STDEV heuristics, comprising analysis of standard deviations (Mednis et al., 2011). This finding suggests that dispersion measures, and standard deviation in particular, are some of the best indicators that can be used to identify road defects based on accelerometer readings.

Another example of using neural networks to assess the road pavement condition based on linear accelerations recorded with an accelerometer has been provided by Ngwangwa et al. (2014). This research paper implies that an implemented neural network is capable of reconstructing both the profile and the typical defects of roads in a manner which ensures high accuracy of mapping in the road pavement condition description. Terzi (2017) describes neural networks used to model the utility coefficient of flexible road pavements. The input data fed in this case into the respective neural network include the parameters of rut slope and depth, as well as comprehensive parameters of road pavement patches and cracks, including the longitudinal ones. A neural network-based decision-making system used when choosing ad-hoc and major repairs has been described by Alsugair and Al-Qudrah (1998). For proper functioning, it requires the network to be trained using a set of data describing the road pavement condition, including parameterised defects which represent it. The data used for the road pavement condition assessment and identification of adequate maintenance procedures were obtained under a comprehensive inspection of a road network section in Saudi Arabia.

Sattar et al. (2018) do a review of the literature of the subject, analysing the available tools and methods used for road infrastructure condition assessment based on the use of a conventional smartphone's accelerometer as an element of the measurement method. The main conclusions arising from this study include the necessity of smoothing the measurement data and the locations, while a perfect solution would be to create algorithms adapting themselves to different vehicle speeds, traffic conditions, vehicle categories, or driving styles. It was suggested that a hybrid approach should be developed, capable of continuously detecting and distinguishing between various road pavement defects using real-time data streaming from smartphone sensors. It is therefore recommended that a free-of-charge multi-platform smartphone application should be developed, one which would be fully automatic and require no user interaction, and operate in the crowdsourcing mode enabling data to be collected via many different road network users.

The aforementioned solutions often encounter implementation challenges which limit their applicability and verifiability. Moreover, it is difficult to clearly define the

accuracy of the implemented models due to the limited availability of the algorithms presented in the paper. Moreover, the unavailability of a database of road measurements or the changes in the road infrastructure condition progressing over time make it impossible to perform comparative analyses.

In many cases, the road surface condition assessment based on thresholds is unclear. Furthermore, methods that use supervised or unsupervised learning require a huge database containing road pavement condition descriptions to train their detection model.

The best approach to the road infrastructure condition assessment seems to be a combination of solutions which enable detection of various road distresses using real-time data crowdsourcing from smartphones, taking into account the condition descriptions contained in the road pavement database obtained with statistical tools. The benchmark approach should rely on a self-adapting and self-learning method. However, this entails the need for a huge database to be created for the sake of identification of adequate relations.

3. System of continuous road pavement condition monitoring

This section of the article presents a system created to assess the condition of road pavements where the measurement procedure performed to describe the road pavement condition is continuous, while the data thus obtained are up-to-date and characterise the area of the transport system being monitored. This solution is based on the concept of telemetric measurements using such technologies as MEMS, GPS, GSM, and GIS, namely making use of the available measuring modules as well as information and communication technologies (ICT) within the space of the smart city being developed. The measuring method covers ordinary road network users, travelling by passenger cars on a daily basis, or drivers of delivery vans and heavy goods vehicles who perform transport services in urban areas. Their participation in the system requires a consent to a method of recording and analysing of data, which they would consider non-invasive, used to describe the motion dynamics of the vehicles in a scope limited to the assessment of road pavement condition.

The proposed concept of a system of continuous monitoring and assessment of the road pavement condition, based on a method of acquisition and analysis of linear acceleration signals, was developed, implemented and tested as one of the tools of the S-mile platform under the international project entitled “smart platform to integrate different freight transport means, manage and foster first and last mile in supply chains” abbreviated to S-mile, implemented under the “sustainable logistics and supply chains” competition within the framework of the ERA-NET Transport III programme. A consortium of six institutions was involved in the project, including businesses and higher technical schools from three countries: Spain (SAITEC, Factor CO₂ and DeustoTech), Turkey (PlusOneMinusOne) and Poland (Faculty of Transport and Aviation Engineering of the Silesian University of Technology).

The assumed concept underlying the development of the road condition tool (RCT) stems from a review of the available solutions for identification of road traffic parameters and road pavement condition assessment as well as from analysis of their comprehensiveness and simplicity of the measurement procedure involved.

3.1. Mobile data logger

One of the tasks performed under the S-mile project was the development and implementation of an operating algorithm for the RCT mobile application which is intended to be installed on mobile devices, either smartphones or tablets, running on the Android operating system's 4.4 release, or higher. In order to make the most of the application's features, the mobile device must have such modules on board as MEMS, GPS, and GSM. The operating algorithm uses the linear accelerations recorded with a high frequency by the MEMS module and combines these data with the GPS-based information on the module location. Measurement data are stored in an output file matching the pre-defined RCT standard and sent to the measurement centre by packet data transmission via the GSM module. The measurement centre should be understood as a local administration unit responsible for the transport system infrastructure, having the RCT server software installed and used for archiving and processing of measurement data.

The features defined for and deployed in the RCT mobile application enable the measurement procedure to be performed in two operating modes: remotely and manually. Remote measuring makes the driver's interference in the mobile device handling redundant, while in the manual mode, the driver is forced to enter such attributes as route identifier or vehicle. However, the choice of the operating mode has no effect on how the vehicle motion dynamics are measured, and so on the measurement procedure as well. The graphical user interface of the application in question has been shown in Fig. 1.

An output file is created as a result of measurements, and it is composed of two parts. One is a heading that contains parameters identifying a number of parameters including the user, the route being travelled, the vehicle and its type, the vehicle weight and payload, as well as the frequency of data acquisition, the latter parameter being crucial from the perspective of identification of the road infrastructure condition. The second part is a series of data of the recorded vehicle motion dynamics, including the follows.

- Measurement time stamp (TIME_STAMP of the EpochTime standard by Mills (1992)).
- GPS coordinates (LAT, LON of the WGS'84 standard).
- Linear accelerations in a three-dimensional co-ordinate system (A_x , A_y , A_z (m/s²)).
- Vehicle speed at the time of measurement (SPEED (km/h)).

Below is a sample fragment of an RCT file containing a series of data describing the vehicle motion dynamics.

The # symbol in Table 1 designates the parameters read from the GPS module, for which the data obtained represent identical values as for the read-out performed in the

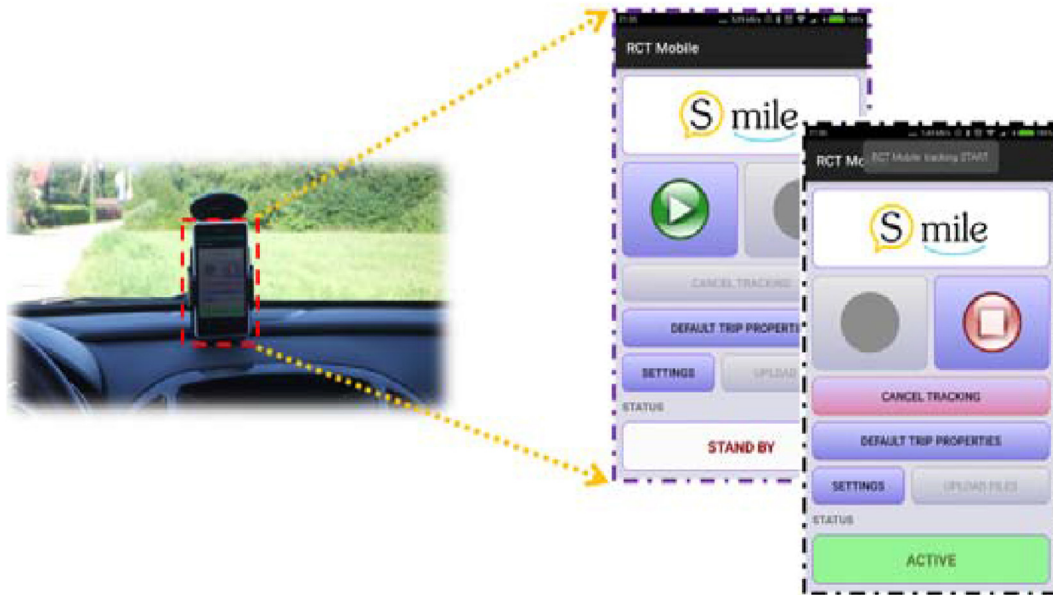


Fig. 1 – Main screen of the RCT mobile application installed on a mobile device.

preceding step. This results from the lower frequency of data extraction from the GSM module compared to the data extracted from the MEMS module as well as the assumed criterion of storage memory usage minimisation. Most typical GPS receivers/modules make it possible to obtain location data with a frequency of up to 1 Hz, which is sufficient for mapping the chosen route on GIS maps. The linear acceleration data extracted from the MEMS module, on the other hand, can be acquired at a higher frequency, which can reach up to 1 kHz for the RCT mobile measuring tool proposed.

3.2. Estimating the road pavement assessment index

The vehicle motion dynamics file is sent by the RCT mobile application to the database, whereby the RCT server confirms having retrieved a complete file with the FTP protocol's acknowledgement signal generated for mobile devices. The RCT server system starts the process of verification and processing of the uploaded data. It checks the continuity of the data series as well as the values of the accelerations and locations obtained according to pre-defined affiliation criteria. Discontinuity of the data series and deviation of

instantaneous values from the mean value trigger the data filtering stage. A cleaned data set is subject to conversion into the GIS system, which means that the route track provided by the GPS is assigned to sections of the OSM's road network. The data conversion process proceeds in stages and requires conformity with such criteria as spatial coverage of the GPS track including the OSM sections, conformity between the GPS track direction and the traffic direction in the OSM sections, and continuity of the GPS track mapping with the route of the OSM sections. However, on account of the different length of the OSM sections and the need to refer the results to the studies conducted by the infrastructure administration agencies using highly specialised equipment, the OSM section was additionally divided into fixed 10 m fragments, excluding the indivisible sections. The data describing the vehicle motion dynamics, aggregated to individual measurement sections, provide a starting point for determining the road network infrastructure condition assessment index.

The road network infrastructure condition assessment index proposed by the author, characterising the road degradation degree as well as pavement defects, was developed with reference to pre-defined technical road classes (GDDKiA, 2015). This index assumes continuous values ranging between

Table 1 – Sample fragment of an RCT file.

TIME_STAMP	LAT	LON	A_x (m/s ²)	A_y (m/s ²)	A_z (m/s ²)	SPEED (km/h)
1503001167048	49.859258	19.023746	−1.1489	9.8932	−3.4529	82.80
1503001167058	#	#	−1.4042	10.3719	−3.2533	#
1503001167068	#	#	−0.8806	10.3512	−3.4874	#
1503001167077	#	#	−0.9581	10.4239	−3.4401	#
1503001167087	#	#	−0.7298	9.9543	−3.2819	#
1503001167097	#	#	−0.9746	9.9085	−3.2613	#
1503001167107	#	#	−0.8151	9.3339	−2.9042	#
1503001167117	#	#	−0.5547	9.7679	−3.2722	#
1503001167126	#	#	−0.7039	9.8145	−3.3597	#
1503001167136	#	#	−0.5040	9.6276	−2.8778	#

0 and 1, where 0 corresponds to a road section of good condition (technical class A), while 1 defines a section of poor condition (technical class D) (GDDKiA, 2015). For the given test road section, the pavement condition assessment index $\delta(d_{\text{sect}})$ is estimated with reference to a single transfer made with a vehicle equipped with a device having the RCT mobile application on board according to the following relationship.

$$\delta(d_{\text{sect}}) = [1 + \exp(-\beta(d - T))]^{-1} \quad (1)$$

where d is sum of absolute differences of the set of accelerations as per Eq. (2), β is parameter defining the above function's curve slope—a high value implies stepwise behaviour of the function, and T is cut-off threshold for the vehicle's free vibrations.

$$d_{\text{sect}} = \sum_j |a_j^y - a_{j-1}^y| \quad (2)$$

where a^y is recorded vertically oriented linear accelerations assigned to the given road section.

A graphical interpretation of the function of road pavement condition description for the given input data based on the results of the measurements performed and the pre-assumed values of parameters β and T has been provided in Fig. 2. Furthermore, the relevant technical road pavement classes as per the guidelines specified in GDDKiA (2015) have also been marked in Fig. 2.

The behaviour of the above function depends on the test vehicle category as well as on its driver's driving style. Having analysed the linear accelerations of vehicles representing different categories against roads of different technical condition, one could assume a constant value of parameter β , corrected depending on the vehicle category. This value was estimated experimentally with reference to the linear accelerations recorded during the measurements performed by more than a dozen drivers of different driving experience in road sections whose accumulated length exceeded 2500 km. What was excluded from the analysis is the data describing the motion dynamics of the vehicles for which the recorded value of linear longitudinal acceleration (coinciding with the traffic direction) was more than 4 m/s^2 , which is described in the literature as braking with partial yet not violent locking of vehicle wheels. Such data typically concerned areas of signal-

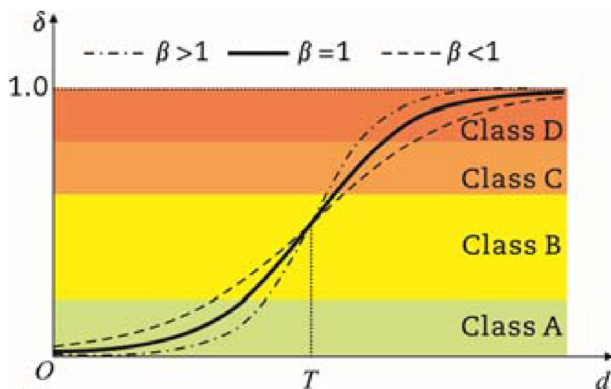


Fig. 2 – Behaviour of the pavement condition assessment function.

controlled intersections and pedestrian crossing approach sections. It was also assumed that the drivers participating in the road infrastructure condition monitoring did not perform any aggressive manoeuvres while driving.

The cut-off threshold for free vibrations of vehicle T depends on the vehicle category, and particularly on its drive-train and power transmission system. The value of this parameter stems from studies of vibrations of vehicles coming to a stop, and it is calculated as a mean value of linear accelerations, i.e., vibrations for the pre-assumed vehicle categories. The procedure for measuring the vehicle's free vibrations is performed once the vehicle has come to a stop, lasting for more than 30 s. In practise, this is usually done while the driver is waiting for an opportunity to enter an intersection, and involves no driver's interference.

Furthermore, in order to take into account different vehicle speed levels, and consequently also the different number of data describing the vehicle motion dynamics for measurement sections of 10 m, the value of the sum of the absolute differences between successive linear accelerations is adjusted by applying the factor of 0.025 of the average vehicle speed over the given measurement section. The reference point is the speed of 40 km/h, assumed as acceptable in a built-up area, while in order to increase the precision of the road pavement condition assessment, the data recorded at speeds $< 20 \text{ km/h}$ are removed from the set.

Fig. 3(a)–(d) provides the graphical interpretation of the relationship between the recorded vehicle's linear accelerations and the estimated road pavement condition assessment index. Fig. 3(a) and (b) pertains to the measurements and the condition assessment index for roads of poor technical condition (class D), classified as critical and requiring immediate ad-hoc repairs or planned major repairs. Fig. 3(c) and (d) covers the measurements and the condition assessment index for roads of satisfactory technical condition (class B), classified as desired, after the surface course of the road pavement was replaced.

Following the assumption that the RCT system being developed operates on a continuous basis in the road network area subject to research, one should take into account the fact that test vehicles can use the same road sections numerous times, and so the procedure to estimate the road pavement condition assessment index for these sections is to be performed multiple times. With an intent to account for all the results obtained, the defects identified, and the repairs performed when determining the road pavement condition assessment index, an arithmetic mean of the assessment indices established for the measurement section was taken into consideration for the following three horizons.

- Time horizon H defined as a period of 30 d preceding the current date.
- Time horizon S defined as a period of 1 d (but at least 10 measurements).
- Time horizon L defined as a period of 1 week (but at least 100 measurements).

The adopted time horizons result from the observation conducted in the initial phase of the S-mile system's

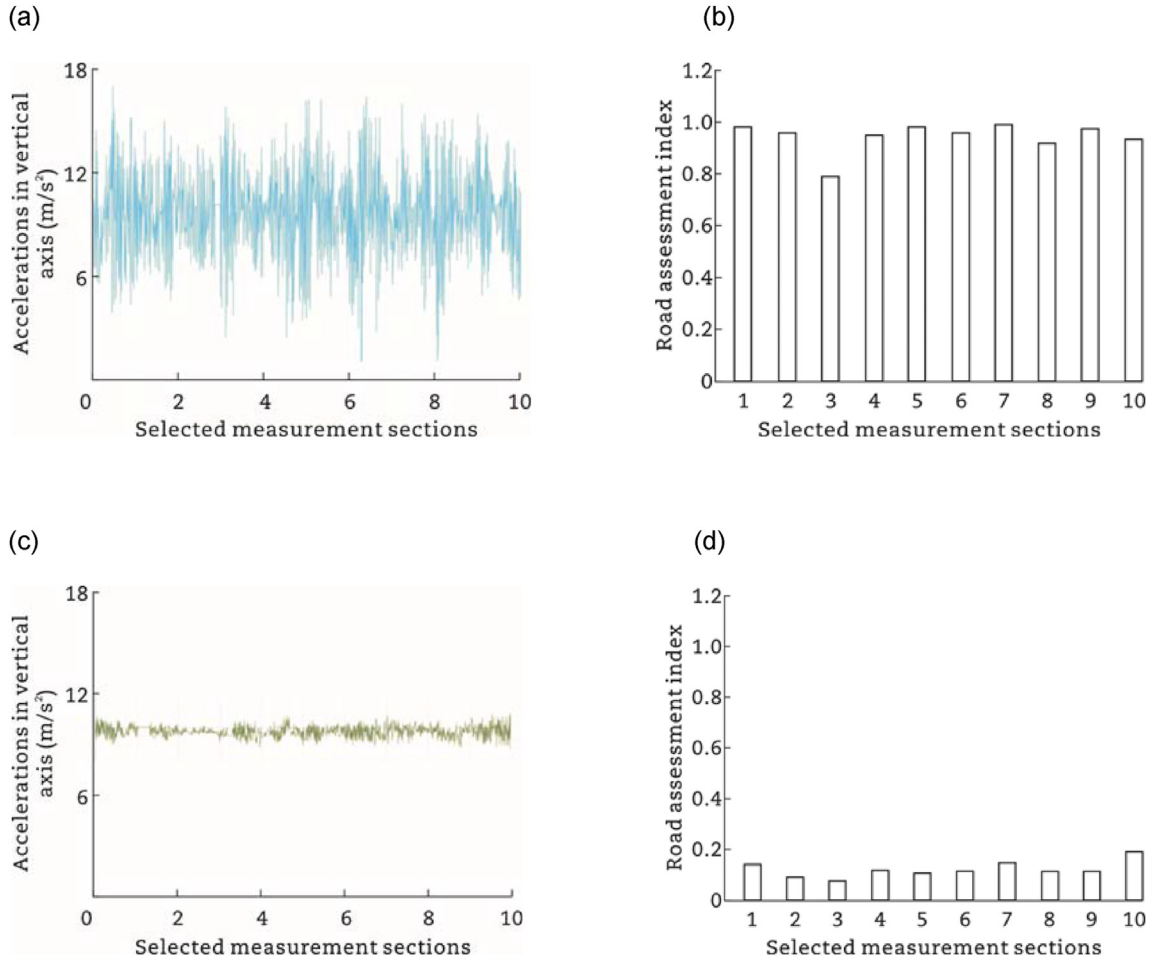


Fig. 3 – Visualisation of the recorded linear accelerations in the vertical axis of the RCT tool with the estimated values of the pavement condition index for selected road sections. (a) Recorded linear acceleration for road of poor technical condition. (b) Estimated pavement condition assessment index for road of poor technical condition. (c) Recorded linear acceleration for road of satisfactory technical condition. (d) Estimated pavement condition assessment index for road of satisfactory technical condition.

implementation in transport companies providing their services in the territory of the Upper Silesian urban area. By increasing the number of users of the RCT tool, the adopted time horizon parameters can be adjusted to shorter periods of time, however, this requires an increase in the number of transfers made in pre-defined measurement sections. It should be noted that the increase in the number of characteristics describing measurement cross-sections exerts a positive impact on the assessment accuracy and provides new information which can be used in new algorithms of the road pavement condition assessment in the future research.

The algorithm used to estimate the mean road pavement condition assessment index for the chosen measurement section based on the set of assessment indices corresponds to the following relationship.

$$\begin{cases} \text{if } |\Delta_{\text{sect}}^L - \Delta_{\text{sect}}^S| \leq 0.12 \\ \text{then } \Delta_{\text{sect}}^H = \sum_i \delta_{\text{sect}}^i / n \text{ for } i = 1, 2, \dots, n \\ \text{else } \Delta_{\text{sect}}^H = \Delta_{\text{sect}}^S \end{cases} \quad (3)$$

where δ_{sect}^i is road pavement condition assessment index estimated of every measurement on the test road section, Δ_{sect}^H is mean road pavement condition assessment index in 30 d time horizon H as in Eq. (3), Δ_{sect}^S is mean road pavement condition assessment index in short time horizon S as in Eq. (4), and Δ_{sect}^L is mean road pavement condition assessment index in long time horizon L as in Eq. (5).

$$\Delta_{\text{sect}}^S = \sum_i \delta_{\text{sect}}^i / n_S \quad \text{for } i = 1, 2, \dots, n_S \quad (4)$$

$$\Delta_{\text{sect}}^L = \sum_i \delta_{\text{sect}}^i / (n_L - n_S) \quad \text{for } i = n_{S+1}, \dots, n_L \quad (5)$$

The proposed solution makes it possible to ensure that the road pavement assessment is kept up-to-date. What it takes into consideration is the deteriorating assessment grade due to the current defects to the infrastructure elements, whose nature of occurrence was sudden and unplanned, but it also ensures improvement of the infrastructure condition once the required minor and major repairs have been performed. This solution requires no interference with the infrastructure

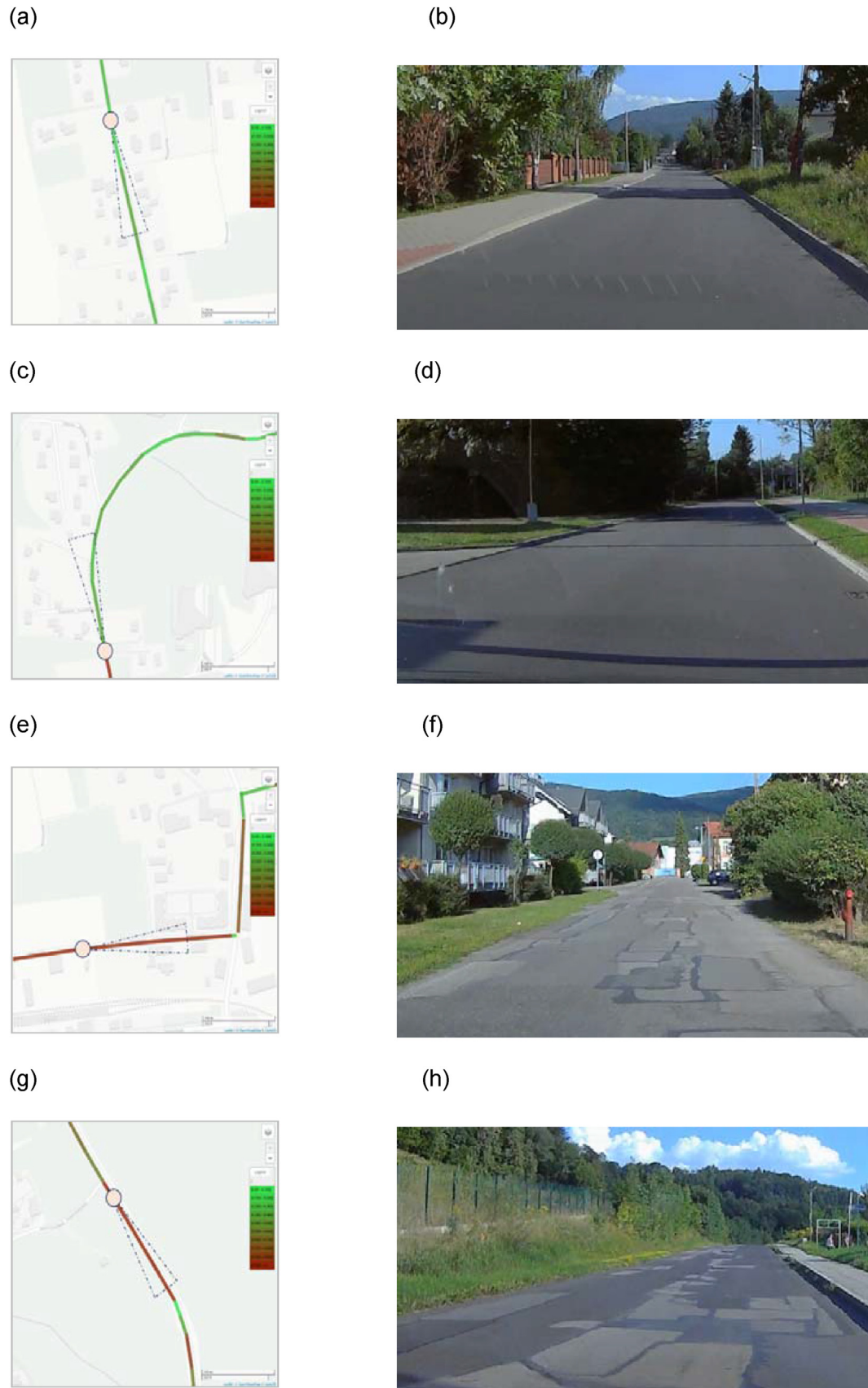


Fig. 4 – Visualization of the road pavement assessments and photos of the studied road. (a) $\delta = 0.253$. (b) Class B–satisfactory condition ($\delta = 0.253$). (c) $\delta = 0.318$. (d) Class B–satisfactory condition ($\delta = 0.318$). (e) $\delta = 0.949$. (f) Class D–poor condition ($\delta = 0.949$). (g) $\delta = 0.910$. (h) Class D–poor condition ($\delta = 0.910$).

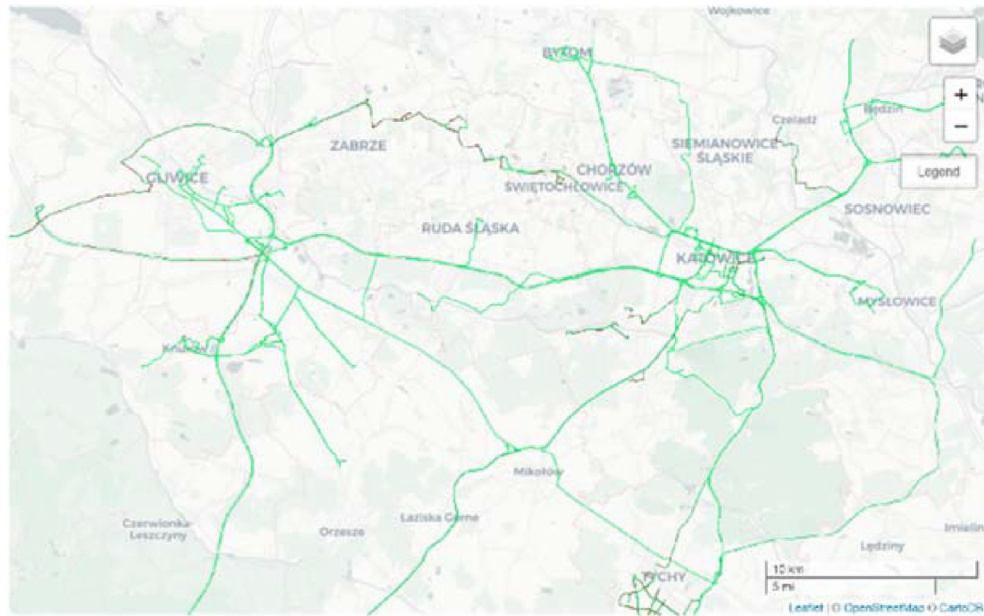


Fig. 5 – Status map of road pavement condition generated with S-mile visualizer tool.

condition monitoring system in order to update the relevant values of the pavement condition assessment indices in selected road network sections.

3.3. Example of infrastructure condition identification

An example of the road pavement condition visualisation with reference to images recorded during measurements has been provided in Fig. 4. Road sections in question are characterised by the mean value of the condition assessment index for the entire measurement section based on measurements performed using the RCT mobile tool. As per the DSN (GDDKiA, 2015) guidelines, sections were categorised, thus obtaining an overall assessment of the road pavement condition.

In order to visualize the acquisition data using the RCT tool, the S-mile visualizer tool was developed as part of the S-mile project. The data processing process takes place in successive steps, after which it is possible to visualize the results fully according to the criteria specified by the user. Data describing the roads are recorded by vehicles in the transport fleet equipped with mobile devices with the RCT mobile application installed and running. Then the acquisition data are received by the RCT server database and visualized (Staniek, 2018b).

The map describing the state of the road infrastructure for the central part of the Silesian Voivodship is presented in Fig. 5. The map shows road sections analogous to those in Fig. 4(a), (c), (e), and (g) but in a large scale of view.

Fig. 6 shows road infrastructure state of Tychy (a city in Silesian Voivodship) for which the measurements carried out by RCT indicate significant differences in the road pavement condition. The obtained results had corresponded to technical condition of road during the measurement period-end of 2018 and beginning of 2019.

4. Evaluation of the road pavement condition monitoring system

In order to evaluate the road pavement monitoring system developed under the study, a series of measurements was performed in the road network of Silesian Voivodship (Poland), thus creating an extensive data set. In previous studies, the correctness of functioning of the RCT tool was roughly verified by comparing the values of condition assessment indices previously obtained with the observed condition of road pavements as well as with the test results obtained from the road infrastructure administration body (GDDKiA, 2015). The repeatability of the data obtained was evaluated with reference to the pre-set test vehicle speed values and the pre-set classes of the vehicles which took part in the measurements. Detailed results of the aforementioned analyses as well as the evaluations obtained and the conclusions stemming from them have been discussed (Staniek, 2018a, 2019).

In the scope covered by this article, it has been proposed that multi-class classifiers should be used for comprehensive assessment of the RCT tool's performance (Powers, 2011). They enable evaluation of the operating efficiency of the road pavement condition assessment algorithm proposed by comparing the results recognised as valid with the results of the road pavement condition assessment studies conducted by the road infrastructure administration body. The administrator's studies were conducted according to the guidelines applied in diagnostics of road pavements in Poland using highly specialised measuring instruments, such as surface analysers and visual devices for automatic assessment of road pavement characteristics (GDDKiA, 2015). The research material for which the operating efficiency evaluation was performed consisted of 3845 actual measurement sections comprising the main roads of Silesian

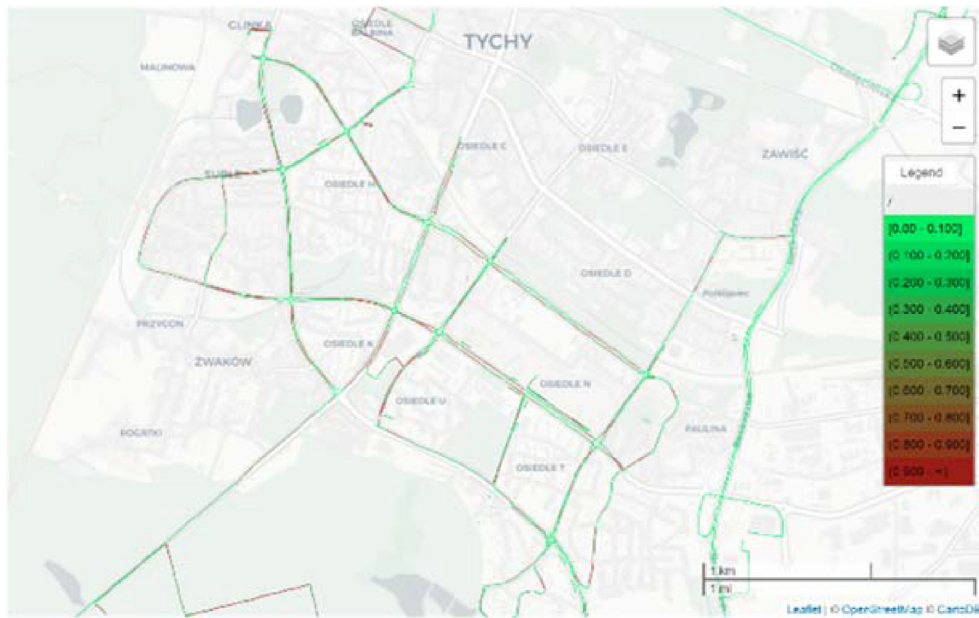


Fig. 6 – Status map of road pavement condition for Tychy.

Voivodship. The results of the infrastructure administrator's studies, representing the identified technical classes, are referred to as reference data, and have been assigned to the pre-defined measurement sections of 10 m in length.

The mean values of the road pavement condition assessment indices established using the RCT tool were classified under pre-defined technical classes (GDDKiA, 2015) according to the relationships provided in Table 2.

The technical classes determining the technical condition of road pavements, those established by means of the RCT tool as well as those provided by the infrastructure administration body, constitute the input data in the analysis of the errors made while classifying the road pavement condition assessments, resulting in creating what is referred to as an error matrix. This matrix provides grounds for evaluation of the RCT tool's efficiency by establishing the values of the following four indices.

- Number of correctly classified conditions of road pavement of the chosen class—TP (true positive).
- Number of incorrectly classified conditions of road pavement of the chosen class—FN (false negative).
- Number of classified conditions of road pavement not assigned to the chosen class (correctly rejected)—TN (true negative).

- Number of classified conditions of road pavement incorrectly assigned to the chosen class—FP (false positive).

The errors made in the classification into the pre-defined road pavement condition assessment classes was analysed using what is referred to as the error matrix (Table 3).

The error matrix is defined in two dimensions, where individual rows correspond to the correct decision classes (reference classes), while columns correspond to the decisions implied by the data classifications based on the mean value of the road pavement condition assessment index.

With reference to the error matrix (Table 3), components TP, FN, FP and TN were established. The values of these components with regard to the pre-defined classes of the road pavement condition description for the measurement sections surveyed have been provided in Table 4.

Having taken components TP, FN, FP and TN into account, one can calculate the measures of efficiency evaluation of the solution proposed. The first of these measures, i.e., accuracy M_A in Eq. (6), determines the probability of correct road pavement condition classification with reference to the mean value of the road pavement condition assessment index.

$$M_A = (TP + TN) / (TP + FN + FP + TN) \quad (6)$$

Table 2 – Classification of road pavement technical condition as per GDDKiA (2015).

Technical condition of road pavement	Assessment index of road pavement condition Δ_{sect}
Class A—good condition	0.0–0.2
Class B—satisfactory condition	0.2–0.6
Class C—unsatisfactory condition	0.6–0.8
Class D—poor condition	0.8–1.0

Table 3 – Error matrix as per proposed continuous road pavement condition monitoring method.

Technical condition of road pavement	Estimated classes by RCT			
	Class A	Class B	Class C	Class D
Class A	1359	152	24	8
Class B	203	1012	89	11
Class C	12	72	376	41
Class D	4	32	29	421

Table 4 – Components of the measures of functional efficiency of the solution proposed.

Technical condition of road pavement	Components of functional efficiency measures			
	TP	FN	FP	TN
Class A	1359	184	219	1809
Class B	1012	303	256	2156
Class C	376	125	142	2792
Class D	421	65	60	2747

The second one, i.e., sensitivity M_S , is a measure of the prediction model's capacity to obtain an appropriate value of the road pavement condition assessment index. It determines the probability of correct classification of the condition chosen from the set of possible ones as shown in Eq. (7).

$$M_S = TP / (TP + FN) \quad (7)$$

Another one, precision M_P , is a measure of accuracy according to which a specific class is predicted with reference to the mean value of the road pavement condition assessment index as shown in Eq. (8). It indicates what fraction of the results determined as positive using the method proposed is actually positive.

$$M_P = TP / (TP + FP) \quad (8)$$

Given the data provided in Table 4 as well as relationships of Eqs. (6)–(8), the following results of the functional efficiency measure were obtained for the continuous road pavement condition monitoring solution in question. Their values have been provided in Table 5.

The continuous road pavement condition monitoring solution proposed under the study, based on the foundation of the analysis of the vehicle motion dynamics, demonstrates the accuracy of 90.5% as well as the sensitivity and precision of 81.7% and 81.5%, respectively. The results thus obtained confirm the high functional efficiency of road pavement monitoring and, consequently, of its condition assessment.

5. Discussions

The RCT tool enables identification and recording of the current technical condition of road infrastructure as well as of the volume and distribution of current freight transport traffic. It is the responsibility of local authorities and road administration bodies to support the process of decision-making in the scope of management and maintenance of road pavement by indicating road sections that require more detailed assessment as well as sections in need of either immediate or planned repairs. This provides survey-based information concerning the assessment of pavement of the entire road network subject to measurements.

From the perspective of transport companies, this tool provides valuable support for planning of freight transport routes by taking the road infrastructure condition into account, i.e., for setting routes which comprise the road network

Table 5 – Evaluation of the functional efficiency of the road pavement condition monitoring system.

Technical condition of road pavement	Functional efficiency measures of RCT (%)		
	M_A	M_S	M_P
Class A	88.7	88.1	86.1
Class B	85.0	77.0	79.8
Class C	92.2	75.0	72.6
Class D	96.2	86.6	87.5
All classes	90.5	81.7	81.5

of good technical condition. It also enables identification of the sections which should be eliminated from the routes planned for the transport services provided. Choosing routes of good technical condition, one can reduce both the probability of cargo damage and the vehicle wear.

The RCT tool developed by the author allows for identification and assessment of the road infrastructure condition on the basis of characteristics describing the vehicle motion dynamics determined with reference to the technical condition of the road used by respective vehicles. The element of key importance is the acquired three-dimensional description of linear accelerations which, when combined with the vehicle speed recorded at the time of readout of the data provided by the MEMS module and of the GPS location, makes it possible to define relationships enabling identification and assessment of the current road infrastructure condition.

The results obtained for the selected road sections confirm the potential of the measurement method proposed. The verification of the RCT tool's functioning and the repeatability of the assessment indices established (Staniek, 2019) confirm that the analysis of linear acceleration signals can be successfully used to identify and describe the condition of road pavement. The studies performed by the author as well as the estimated values of the indices in question (Staniek, 2018a) have implied that the road pavement condition assessment measures are independent of the running speed of the test vehicles performing the measurements as well as of their category. An exception to the foregoing regularity is the category of heavy goods vehicles, for which the method proposed should evolve further on towards designing a model of an adequate driver's cab shock absorbing system.

The main limitation of the RCT tool developed under the research is the need for updating the parameters of Eq. (1), which allows for the waveform to be smoothed out. In order to eliminate this problem, one must create a huge database and determine parameters for each vehicle class. It should be added that the vehicle drivers and their driving styles can affect the accuracy of the road surface condition description. However, by identifying drivers showing a very dynamic driving style in the entire data set, it is possible to reject such runs using a typical statistical model for accelerations along the vehicle.

The RCT tool creates an enormous data set describing vehicle motion dynamics against the road network, which—taking both the current and the archival data into account—can be classified as big data. Besides the main function of the tool in question, namely road pavement

condition assessment, the data series it records enable determination of other traffic parameters, such as travelling speed or congestion. It also allows for a vehicle to be diagnosed in operational terms, i.e., with regard to the operation of its powertrain, power transmission or suspension system. Not only does it allow for the assessment of the driver's behaviour in terms of how dynamic one's manoeuvres are, how safely one drives, and what travelling comfort one ensures, but even of the driving style against the travelling cost-effectiveness. However, the RCT mobile APP excludes groups of transport system users without Android OS from the road pavement condition monitoring. However, it is possible that the RCT APP in question be deployed on other operating systems (e.g., iOS for iPhone, iPod touch or iPad mobile devices) as an alternative.

What should be emphasised is the possibility to introduce descriptors of the travel's impact on the transport system condition. Besides the indicators of climate change, public health impact and noise emission, which are commonly being determined at present, one can add new measures to describe in detail the impact of individual compounds of the harmful substances produced by vehicle traffic. This thematic area will be in focus of further research on the use of the RCT system in the smart city space.

Assuming a vision that the system developed by the author is to be implemented as a smart city element based on all users of the transport system, the local authorities can benefit from a significant increase in the number and scope of valid data describing actual traffic conditions in the road network which can be used, among other purposes, in traffic control and management systems. Without direct involvement, the system users support the functioning of the smart city, and in return they receive the support they need to efficiently plan their travels and to improve travelling comfort. The reluctance to share information can be overcome by using promotions, privileges or incentives for the transport system users using the solution in question, managed by the city's local authorities.

6. Conclusions

The road pavement condition assessment tool developed and implemented by the author is a solution referred to as road condition tool, whose measuring procedure is performed on a continuous basis by vehicles using the road network. The tool enables identification and assessment of the current condition of road pavements as well as of traffic volume and distribution with regard to freight transport. Once implemented, the RCT tool allows for identification and assessment of the road infrastructure condition on the basis of characteristics describing the vehicle motion dynamics determined with reference to the technical condition of roads. The key element of the system in question is the three-dimensional description of linear accelerations which—when combined with the vehicle running speed—makes it possible to assess the current road condition, and—when complemented with geolocation data—enables parameterisation and keeping records of the road network defects. Having successfully verified the RCT tool's functionality and the repeatability of the assessment

indices thus obtained, one can confirm that the analysis of linear accelerations can be effectively used to evaluate the condition of road pavement. Furthermore, once implemented, the tool in question does not generate any hindrance to other traffic participants. It may function on a continuous basis while transport services are performed within city or agglomeration borders, and yet it needs no intervention from the driver.

Conflict of interest

The author does not have any conflict of interest with other entities or researchers.

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